AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M348
Applied Statistical Modelling

Midterm Examination (MTA)
Spring Semester 2024/2025

Date: 08/April/2025

Number of Exam Pages: 5 (including this cover sheet)	**Time Allowed:** 2 Hours

Instructions:

- Total Marks: 30
- This exam consists of **three** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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PART 1: True or False Questions [5 x 1=5 marks]

Which of the following statements are true/false?

a) The TSS (Total Sum of Squares) is used to measure the total variation in the response variable from its predicted values.
b) Non-parametric models, do not assume a specific form for the underlying data distribution.
c) In multiple regression, the backward selection method starts with a model that includes all predictors and removes the least significant ones, not a model with no predictors.
d) In multiple regression, collinearity between predictor variables can inflate the standard errors of the coefficients, leading to unreliable estimates.
e) A non-flexible model like linear regression will generally perform better than flexible models when the true relationship in the data is linear.

PART 2: Simple Linear Regression [11 x 1= 11 marks]

ALL QUESTIONS ARE REQUIRED

The olympic dataset contains information about nations and their performance in the Summer Olympics for the years 1996, 2000, 2004, 2008, 2012, and 2016. Specifically, the dataset includes the following variables:

Medals: The total number of medals won by a nation in a given Olympic year, with the data collected as of 9 August 2021, the day after the Tokyo 2020 Olympics concluded.
Population: The population size (in millions) of a nation in the year of the previous Olympics.
For this task, you are asked to fit a simple linear regression model where the response variable is medals and the explanatory variable is population. This model aims to explore the relationship between the population size of a nation and the number of medals it won at the Summer Olympics.

```
Residuals:
    Min      1Q  Median      3Q     Max
-57.952  -3.744  -3.479  -0.948 103.529

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 3.463931   0.384820   9.001   <2e-16 ***
population  0.044627   0.002882  15.484   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 12.16 on 1067 degrees of freedom
Multiple R-squared:  0.1835,    Adjusted R-squared:  0.1827
F-statistic: 239.8 on 1 and 1067 DF,  p-value: < 2.2e-16
```


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1. Complete the code below to fit the simple linear regression model with response variable medals and explanatory variables population, and then display a summary of the results from fitting the model.
    a. modelFit =lm(__~__)
    b. summary(__)
2. Using the results obtained in part (1), what is the fitted equation for this model?
3. Interpret the coefficient of the population variable. What does it mean in the context of predicting the number of medals?
4. What does the intercept of 3.463931 represent in this model? Is it meaningful in this context?
5. What does the multiple R^2 value indicate about the model's performance?
6. Why is the p-value of the population coefficient important? What does it tell us about the significance of the relationship?
7. What is the null hypothesis for the coefficient of population in this regression model? Based on the results, should we reject or fail to reject the null hypothesis?
8. How would you interpret the residual standard error of 12.16?
9. If a country has a population of 50 million, how many medals does the model predict it will win?
10. How would you assess whether this model is a good fit for predicting the number of medals?
11. What assumptions of simple linear regression should be checked before trusting the model results?

PART 3: Multiple Linear Regression [14 marks]

ALL QUESTIONS ARE REQUIRED

Fit a multiple linear regression model to explore the relationship between the total number of medals won by a nation at the Summer Olympics and several explanatory variables. The response variable, medals, represents the total number of medals won by a nation in a given Olympic year, with the data collected as of 9 August 2021 (the day after the Tokyo 2020 Olympics concluded). The explanatory variables include:

- lagMedals: Number of medals won by the nation in the previous Olympic Games.
- Host: Binary variable (1 or 0) indicating whether the nation hosted the previous Olympics.
- nextHost: Binary variable (1 or 0) indicating whether the nation will host the next Olympics.
- Population: Total population size (in millions) of the nation in the year of the previous Olympics.
- logGdp: Logarithm of the nation's GDP per capita (in thousands of US dollars) in the year of the previous Olympics.

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- logPopulation: Logarithm of the nation's population size (in millions) during the previous Olympic cycle

```
Residuals:
    Min      1Q  Median      3Q     Max
-25.0166  -0.5198  -0.0981   0.3078  29.2117

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  -0.2463129  0.1704148  -1.445  0.14865
lagMedals     0.9320374  0.0090688 102.774  < 2e-16 ***
host         12.8327892  1.3103221   9.794  < 2e-16 ***
nextHost      8.6128795  1.2870078   6.692 3.55e-11 ***
population    0.0015874  0.0008636   1.838  0.06631 .
logGdp        0.1943544  0.0697709   2.786  0.00544 **
logPopulation 0.1485136  0.0537330   2.764  0.00581 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.09 on 1062 degrees of freedom
Multiple R-squared:  0.9475,    Adjusted R-squared:  0.9472
F-statistic:  3196 on 6 and 1062 DF,  p-value: < 2.2e-16
```

1. Why is it beneficial to use multiple regression instead of simple regression in this case? (1 pt)
2. What is the interpretation of the coefficient for host (12.83)? (1 pt)
3. What does the adjusted R^2 value tell us, and why is it preferable to multiple R^2 when comparing models? (1 pt)
4. Which explanatory variables have the strongest effects on the number of medals won, based on their coefficients and statistical significance. (1 pt)
5. The residual standard error is 3.09, while in the simple regression model, it was 12.16. What does this tell us about model improvement? (1 pt)
6. How could multicollinearity affect the results of this model? Provide one technique that could be used to address it. (1.5 pts)
7. What steps would you take to determine if this model is overfitting the data? (1 pt)
8. What is the trade-off between model complexity (flexibility) and interpretability in regression modeling? (1.5 pt)
9. In the empty cell below, apply both forward and backward stepwise regression, starting from the full model. Use the following variables: lagMedals, population, gdp, logPopulation, logGdp, host, nextHost, and year. The goal is to find a parsimonious model that includes some or all of these variables. Complete the following code: (1.5 pt)

```
step(??, direction="both", scope=list(lower=??, upper=??))
```

10. What is the purpose of using stepwise regression in model selection? (1 pt)

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11. To find the best model, should we minimize or maximize the AIC? Explain based on the results from the stepwise regression. (1 pt)
12. Based on the results of the stepwise regression, what variables are included in the final model? Explain (1.5 pts)

```
Start:  AIC=2426.72
medals ~ lagMedals + population + gdp + host + nextHost + year

             Df Sum of Sq    RSS    AIC
- year        1         7  10221 2425.5
<none>                     10214 2426.7
- gdp         1        32  10245 2428.1
- population  1        63  10277 2431.3
- nextHost    1       445  10658 2470.3
- host        1       912  11125 2516.1
- lagMedals   1    124708 134921 5183.8

Step:  AIC=2425.49
medals ~ lagMedals + population + gdp + host + nextHost

             Df Sum of Sq    RSS    AIC
<none>                     10221 2425.5
- gdp         1        30  10251 2426.6
+ year        1         7  10214 2426.7
- population  1        62  10283 2430.0
- nextHost    1       446  10667 2469.1
- host        1       914  11134 2515.0
- lagMedals   1    124732 134953 5182.0
```

End of questions

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